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Plasma Proteomic Markers of Interleukin-1β Pathway Associated with Incident Age-Related Macular Degeneration in Persons with AIDS

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Objective: To evaluate the associations of plasma inflammatory proteins with age-related macular degeneration (AMD) in persons with the AIDS, using a discovery-based proteomics approach.

Design: A nested case-control study (analysis 1) and nested cohort study (analysis 2).

Participants: Persons with AIDS enrolled in the Longitudinal Study of the Ocular Complications with AIDS (LSOCA).

Methods: Cryopreserved plasma specimens obtained at baseline were assayed for inflammatory proteins using the Olink Inflammation Explore Panel 1. In analysis 1, baseline proteomic profiles for 26 persons with AIDS and incident intermediate-stage AMD 5 to 10 years after baseline and 49 matched controls (matched for age, biologic sex, race/ethnicity, and follow-up) without AMD were compared. In analysis 2, 475 persons from LSOCA with baseline plasma inflammatory proteomic profile measurements were followed for incident cataract and mortality.

Main Outcome Measures: Incident intermediate-stage AMD; incident cataract; and mortality.

Results: Of 365 measurable plasma inflammatory proteins, 118 (32%) were associated with incident intermediate-stage AMD at the false discovery rate-adjusted $Q < 0.05$ level after adjustment for smoking, CD4+ T count, and plasma human immunodeficiency virus RNA level. Gene ontology pathway enrichment analysis identified the interleukin (IL)-1β pathway and wound healing pathways, including tissue inhibitor of metalloproteinase 3, as significantly associated with incident AMD. These associations were qualitatively different from those associated with incident cataracts, where elevated levels of inflammatory proteins were associated with a decreased risk of cataracts. A much broader number of inflammatory pathways, including those related to the adaptive immune system, were associated with mortality.

Conclusions: Upregulation of the IL-1β pathway appears to be associated with an increased risk of incident AMD in persons with AIDS. Given the availability of inhibitors of this pathway, inhibition of the IL-1β pathway may provide a therapeutic avenue for treatment of AMD.

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Several lines of evidence have implicated inflammation and the immune system, particularly the innate immune system, in the pathogenesis of age-related macular degeneration (AMD).[1–3] Elevated blood levels of C-reactive protein are associated with the subsequent development of AMD, and patients with late-stage AMD have elevated levels of activated monocytes circulating in the blood.[1,2] Variants in several complement genes are associated with AMD, as are variants in tissue inhibitor of metalloproteinase 3 (TIMP3), a regulator of tissue metalloproteinases involved in tissue remodeling.[3,4]

Modern antiretroviral therapy (ART) of human immunodeficiency virus (HIV)-infected persons results in immune recovery and substantially improves the lifespan of HIV-infected persons versus that seen in the era prior to modern ART.[5–8] Nevertheless, ART-treated, immunorestored, HIV-infected persons exhibit a phenotype characterized by features of accelerated and accentuated aging and have a shorter lifespan than that of comparably aged HIV-uninfected persons.[9–11] Both HIV-uninfected older persons and ART-treated, immunorestored HIV-infected persons exhibit a similar immunologic profile, characterized by chronic immune system activation and systemic inflammation, which is associated with mortality in both HIV-infected and HIV-uninfected persons.[12–16] This accelerated/accentuated aging in ART-treated HIV-infected persons, coupled with their similar immunological features to older HIV-uninfected persons, provides an opportunity to explore the role of inflammation in the pathogenesis of age-related diseases in a population with accelerated/accentuated age-related diseases.

Compared to HIV-uninfected persons, people with the AIDS have an increased age-adjusted and sex-adjusted

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prevalence of intermediate-stage AMD,[17] an increased race/ethnicity-adjusted and sex-adjusted incidence of intermediate-stage AMD,[18] an increased prevalence of cataract at younger ages, and an increased incidence of cataract surgery.[19] In persons with AIDS, hypothesis-driven evaluation of selected plasma biomarkers suggested that incident AMD may be associated with activation of the innate immune system, whereas incident cataract was not associated with elevation of these biomarkers.[20,21] We employed a discovery-based, proteomic approach to perform a comprehensive evaluation of the relationship between inflammatory plasma proteomic levels and AMD.

Methods

This study used cryopreserved plasma specimens from the Longitudinal Study of Ocular Complications of AIDS (LSOCA) cohort, a prospective observational study of participants with AIDS, all of whom received ART.[22] Enrollment, follow-up, examination details, imaging and laboratory testing procedures, and characteristics of the cohort have been described previously.[17,18,22] Briefly, there were 1825 participants with baseline photographs and without ocular opportunistic infections for evaluation of prevalent AMD at enrollment and 730 participants without AMD at enrollment with 5- and/or 10-year follow-up retinal photographs available for evaluation of incident AMD.[17,18] Participants were evaluated for the presence of intermediate-stage AMD (Age-Related Eye Disease Study stage 3) or greater from photographs taken at enrollment (baseline) and at 5- and 10-year follow-up visits by graders at the Wisconsin Reading Center, masked to clinical data, as previously described.[17,18,20,21,23,24] The presence of cataract was graded at every study visit on slit lamp examination, using standard semiquantitative grading systems, and incident cataract was defined as any of the following in either eye: (1) ≥1+ nuclear sclerosis; (2) ≥1+ cortical spoking; (3) ≥1+ posterior subcapsular cataract; or (4) new-onset pseudophakia or aphakia in a participant without cataracts at enrollment, as previously described.[19–21] The Longitudinal Study of Ocular Complications of AIDS adhered to the principles of the Declaration of Helsinki and was conducted with institutional board approval at each clinical and resource center. All participants gave written informed consent.

Proteomic Analyses

Cryopreserved enrollment plasma specimens were thawed and assayed for inflammatory proteins using the Olink Inflammation Explore panel, a high-throughput proximity extension assay proteomics platform,[25] consisting of 384 plasma proteins involved in inflammation or immunoregulation. Of the 384 proteins, 365 passed quality control requirements and were included in the current analysis. Relative concentration for each protein (relative to all other samples assessed in the study) is expressed in normalized protein expression units on a $\log_2$ scale, with each unit increase corresponding to a relative twofold increase in a given protein's concentration.

Evaluation of Baseline Plasma Inflammatory Proteins and Incident AMD

Participants without AMD at enrollment but with incident AMD during follow-up and at least 1 matched control (who had not developed AMD at the time a case developed AMD, individually matched on age [by decade of age], biologic sex, and race/ethnicity) from LSOCA were selected for a nested case-control study of baseline inflammatory proteins and incident AMD. The relationship between plasma proteins and incident AMD was assessed with conditional logistic regression, adjusted for current smoking status, CD4+ T cell count, and $\log_{10}$ plasma HIV RNA level.

Evaluation of Baseline Inflammatory Proteins and Incident Cataract and Mortality

In an effort to evaluate if the relationship between inflammatory proteins and incident AMD differed from the proteins that are associated with other aging-related ocular diseases or overall mortality, we leveraged the baseline samples used to assess the factors associated with baseline prevalent AMD in the cohort. Repurposing this nonoverlapping subset of 475 participants from LSOCA, we analyzed the relationship between baseline proteomics and both incident cataract and mortality. All sub-hazard ratios were adjusted for age, sex, race/ethnicity, enrollment AMD status, and glucocorticoid use. For all proteomics analyses, the Benjamini–Hochberg method was used to correct for the false discovery rate (FDR), and FDR-adjusted Q values < 0.05 were considered statistically significant.[26] Known biologic relationships between statistically significant proteins ($Q < 0.05$) were plotted using STRING network maps (https://string-db.org/).[27] Pathway enrichment analysis was performed using gene ontology, restricting to annotated pathways including the proteins represented in the Olink panel and further restricting to immunologic pathways given the scope of our hypotheses.[28] All data analyses were conducted using R (version 4.2.1).

Results

The characteristics of the study populations have been described in detail previously but comprised people with advanced AIDS with a median age of 44 years, all of whom received ART during follow-up.[20,21] For the nested case-control analysis of incident AMD, plasma samples from the enrollment visit were sampled from 26 cases who subsequently developed incident intermediate-stage AMD and 49 matched controls without AMD. For the study assessing the baseline plasma inflammatory proteins associated with incident cataract and with mortality, baseline plasma samples from a cohort of 475 participants enrolled in LSOCA were used.

Plasma Inflammatory Proteins and Incident AMD

In the adjusted matched case-control analysis, a total of 118 inflammatory proteins were significantly associated with incident AMD at $Q < 0.05$. As evident in the volcano plot (Fig 1A, Table S1), all but one (leucine rich repeat neuronal 1) of the proteins with Q-value <0.01 were associated with an increased risk of AMD (Fig 1A), suggesting a very strong and consistent association between inflammation and incident AMD in this setting. As many of these inflammatory proteins are members of linked pathways in vivo, we plotted those known protein interactions using a protein network map (Fig 1B). As expected, many of the inflammatory proteins significantly associated with AMD had multiple known biologic interactions with each other. Among the significant proteins was TIMP3 (adjusted hazard ratio: 1.69 per twofold increase,

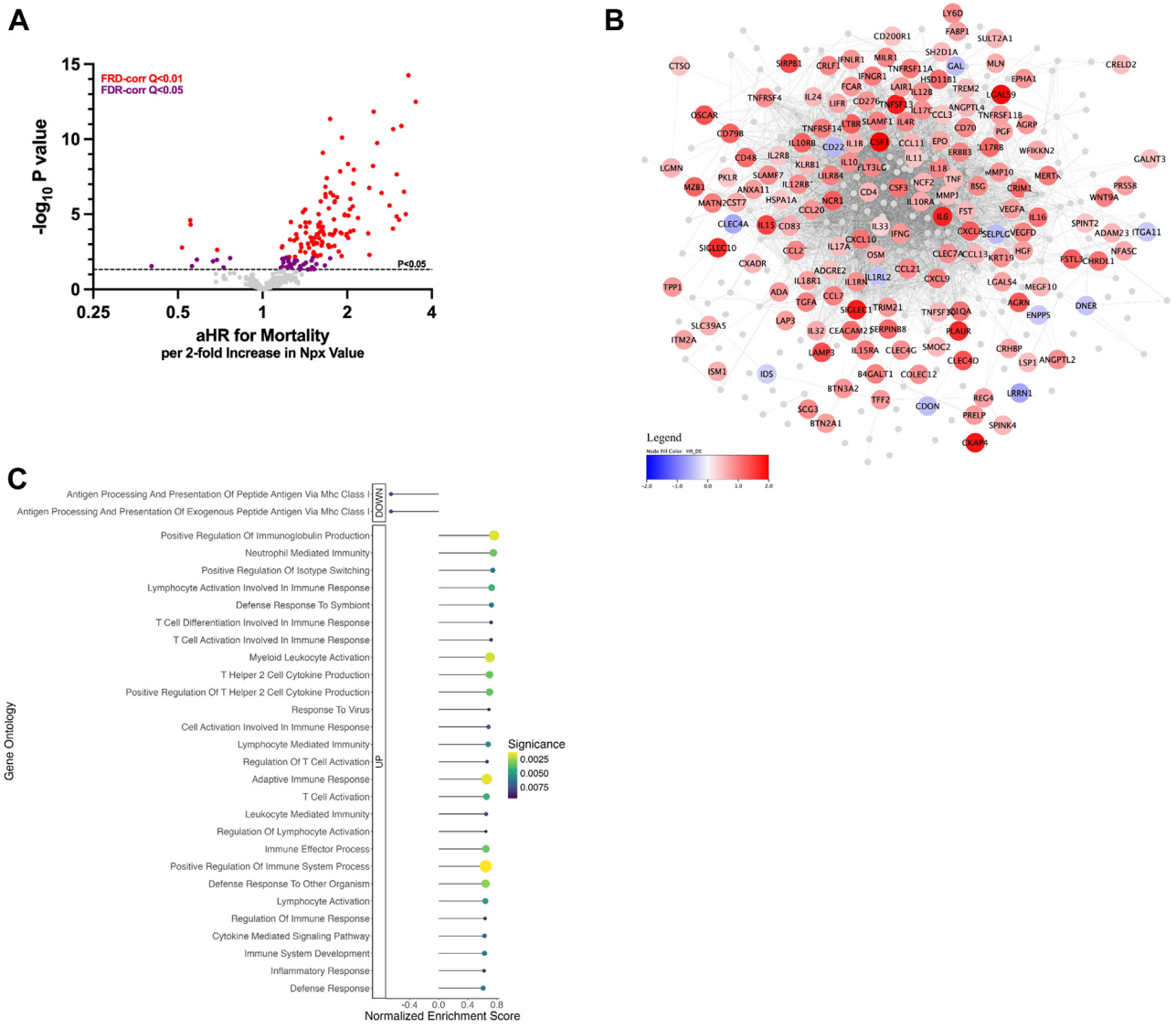


Figure 1. Plasma inflammatory proteins and the IL-1β pathway associated with incident AMD in treated HIV. **A,** The negative $\log_{10}$ *P* value is plotted by the adjusted odds of AMD for each twofold increase in Npx value for each plasma protein in the Olink Inflammation Explore panel. Benjamini–Hochberg FDR Q values < 0.05 are colored in purple, and <0.01 in red. **B,** The protein network map (using the STRING database) of all proteins significantly associated with AMD (Q < 0.05) is shown, labeling each significant protein with color depicting the adjusted odds ratio (positive association in red, negative association in blue); nonsignificant proteins are unlabeled in light gray. **C,** Proteomic pathways associated with incident AMD by gene ontology pathway enrichment analysis are depicted, with the normalized enrichment score on the x-axis and significance level denoted by color. **D,** The adjusted odds ratios for all proteins represented in the IL-1β pathway are plotted in a protein network map, with the color of each dot denoting the odds ratio and the size representing the significance levels. Proteins significant at the Q < 0.05 level are circled in green. AMD = age-related macular degeneration; FDR = false discovery rate; HIV = human immunodeficiency virus; IL = interleukin; NPx = normalized protein expression.

Q = 0.013), a matrix remodeling protein, genetic variants of which are associated with AMD in the general population.[4] No other proteins from genes in which variants are known to be associated with AMD were represented in the Olink Inflammation I panel. Gene ontology pathway enrichment analysis was used to assess in an unbiased way whether any known annotated pathways, including proteins represented in the Olink panel, were enriched among those with incident AMD. Although several pathways associated with wound healing were increased in those with incident AMD, negative regulation (or suppression) of the interleukin (IL)-1β pathway was associated with a decreased risk of AMD. While this gene set enrichment analysis was based on just the 3 proteins represented in these pathways, this finding, along with prior research from our group and others, led us to directly assess the relationship between all proteins in the IL-1β pathway represented in the Olink panel and incident AMD in a dedicated protein network map. Consistent with the unbiased pathway enrichment analysis, 14 of 43 total proteins in the IL-1β pathway—including IL-1β itself

(adjusted hazard ratio 1.91 per twofold increase, Q = 0.0075)—were associated with an increased risk of AMD at the FDR-corrected Q < 0.05 level.

Plasma Inflammatory Proteins and Incident Cataract

To address the specificity of this link between inflammation and the IL-1β pathway and AMD, we assessed whether the same inflammatory markers were associated with a different age-related eye disease, namely incident cataract. Sub-hazard ratios were assessed for cataracts, accounting for the competing risk of mortality. In the adjusted analysis and after accounting for the competing risk of mortality, 131 inflammatory proteins were significantly associated with incident cataract (FDR-corrected Q < 0.05, Fig 2A, Table S2). Of these, the vast majority (87%) were associated with a decreased risk of incident cataract, a qualitatively different pattern than that observed for incident AMD. Indeed, many of the same proteins that were associated with an increased risk of AMD were associated with a decreased risk of cataract (Fig 2B). Gene ontology pathway enrichment analysis also identified only a few pathways related to antigen processing and presentation

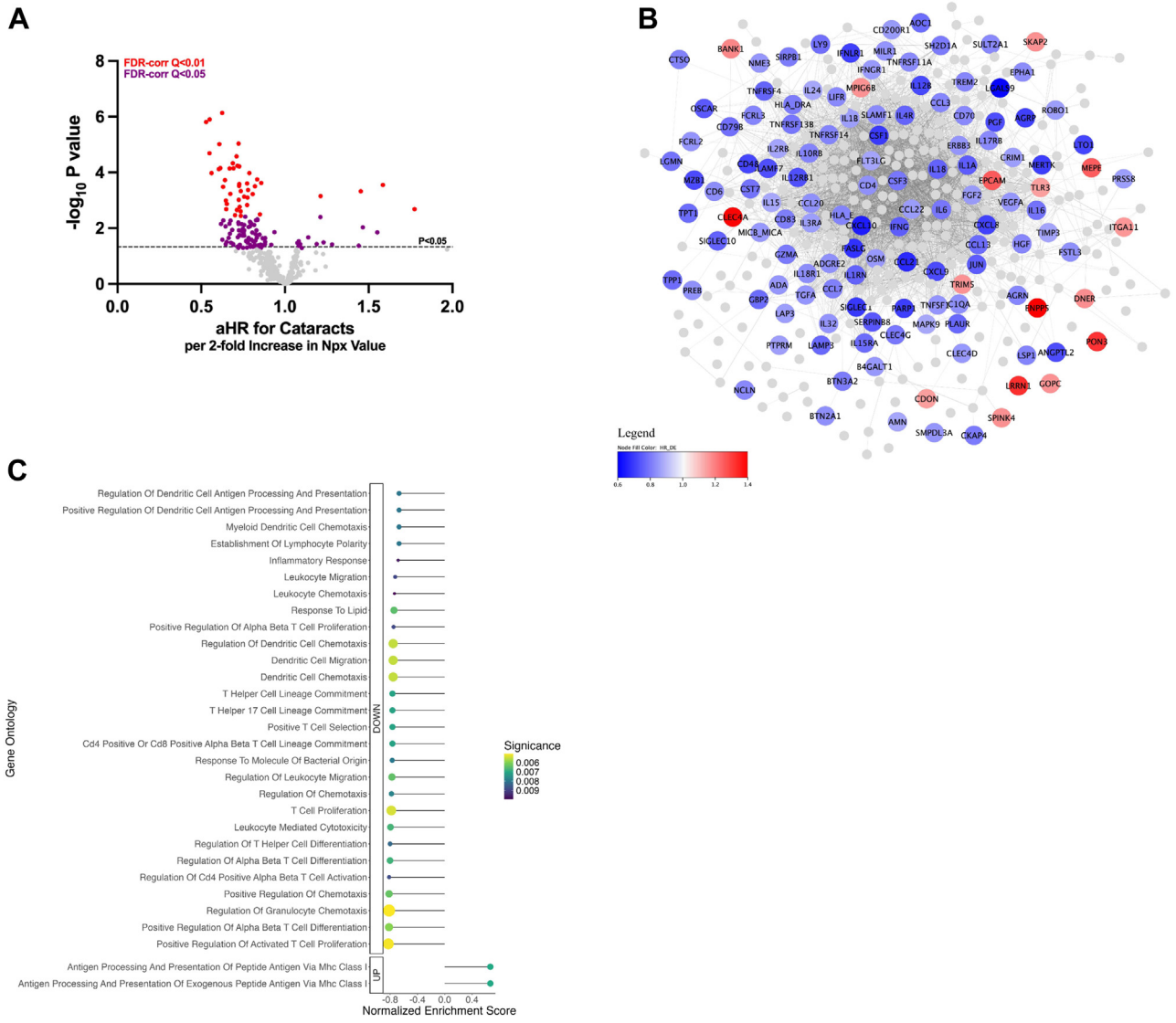


Figure 2. Plasma inflammatory proteins associated with incident cataracts in treated HIV. **A,** The negative $\log_{10}$ *P* value is plotted by the adjusted sub-hazard ratio for cataracts (i.e., after accounting for competing risk of mortality) for each twofold increase in Npx value for each plasma protein in the Olink Inflammation Explore panel. Benjamini–Hochberg FDR Q values < 0.05 are colored in purple, and <0.01 in red. **B,** The protein network map (using the STRING database) of all proteins significantly associated with cataracts (Q < 0.05) is shown, labeling each significant protein with a color depicting the aHR (positive association in red, negative association in blue); nonsignificant proteins are unlabeled in light gray. **C,** Proteomic pathways associated with incident cataracts by gene ontology pathway enrichment analysis are depicted, with the normalized enrichment score on the x-axis and significance level denoted by color. aHR = adjusted hazard ratio; FDR = false discovery rate; HIV = human immunodeficiency virus; NPx = normalized protein expression.

associated with an increased risk of cataract (Fig 2C). The vast majority of identified inflammatory pathways—spanning innate and adaptive immune cell activation and chemotaxis—were associated with a decreased risk of cataract.

Plasma Inflammatory Proteins and Mortality

We assessed whether the AMD-associated inflammatory pathways were simply reflective of the pathways associated with mortality risk or a more specific subset of those pathways. Over 10 years of follow-up, 24% of the 475 participants died. After adjustment, higher levels of 146 proteins were associated with increased mortality risk, and 11 proteins were associated with decreased mortality risk ($Q < 0.05$, Fig 3A, Table S3). Many of the same proteins associated with AMD risk also predicted an increased risk of mortality (Fig 3B), but a much broader array of inflammatory pathways was associated with increased mortality than for AMD. For example, when excluding proteins that also predicted AMD, many pathways associated with lymphocyte and adaptive immune activation continued to be predictive of mortality (Fig S1). While the IL-1β and wound healing pathways were the

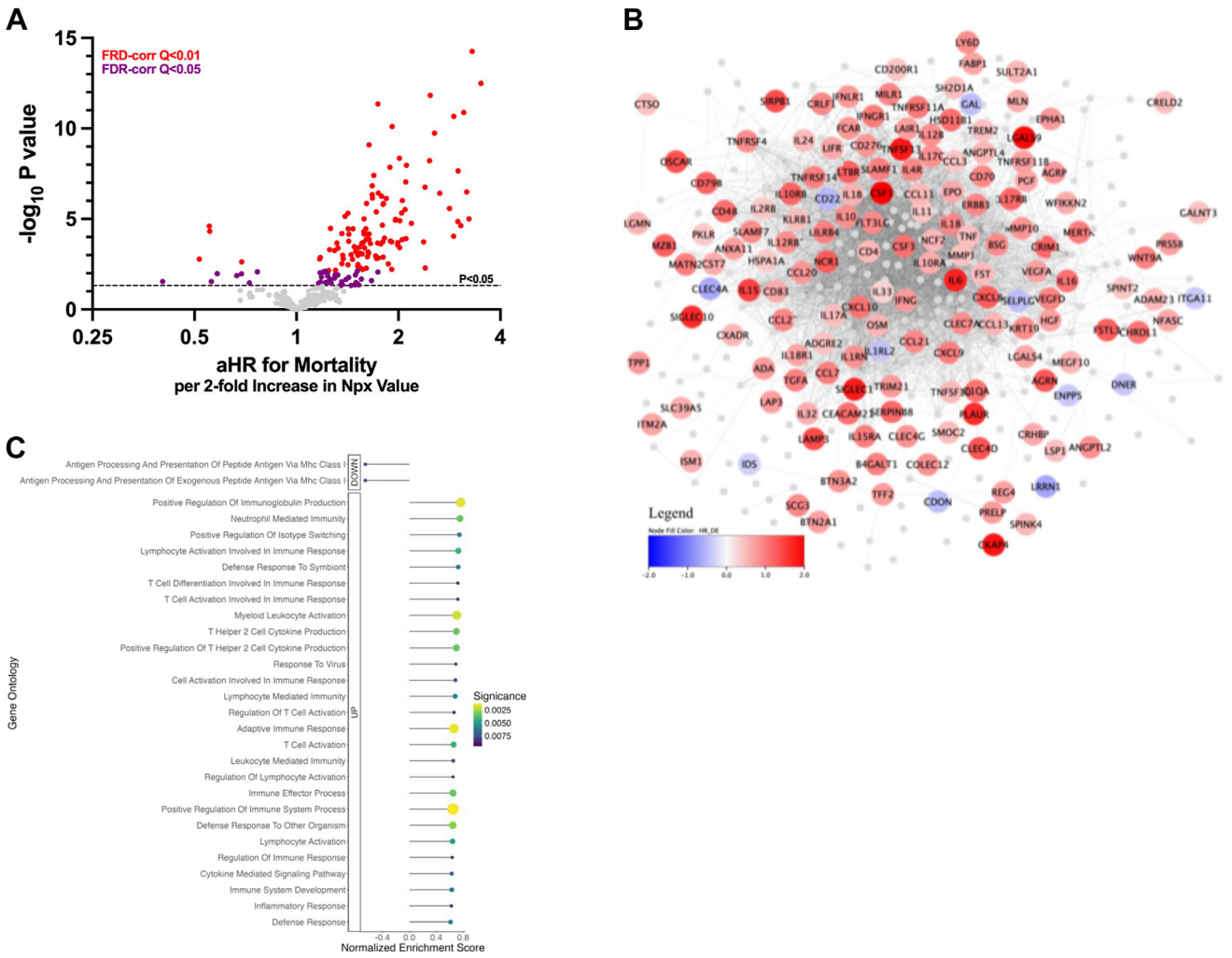


Figure 3. Plasma inflammatory proteins associated with incident mortality in treated HIV. **A,** The negative $\log_{10}$ *P* value is plotted by the aHR for mortality for each twofold increase in Npx value for each plasma protein in the Olink Inflammation Explore panel. Benjamini–Hochberg FDR Q values <0.05 are colored in purple, and <0.01 in red. **B,** The protein network map (using the STRING database) of all proteins significantly associated with mortality ($Q < 0.05$) are shown, labeling each significant protein with a color depicting the aHR (positive association in red, negative association in blue); nonsignificant proteins are unlabeled in light gray. **C,** Proteomic pathways associated with incident mortality by gene ontology pathway enrichment analysis are depicted, with the normalized enrichment score on the x-axis and significance level denoted by color. aHR = adjusted hazard ratio; FDR = false discovery rate; HIV = human immunodeficiency virus; NPx = normalized protein expression.

primary identified pathways associated with AMD risk, a far broader collection of pathways spanning both the innate and adaptive immune systems, leukocyte activation, and chemotaxis were associated with an increased risk of mortality (Fig 3C).

Discussion

Previous data have shown that persons with AIDS have an increased age- and sex-adjusted prevalence and an increased sex- and race/ethnicity-adjusted incidence of intermediate-stage AMD.[17,18] In this population, hypothesis-driven experiments demonstrated that incident AMD was associated with increased baseline plasma levels of C-reactive protein, soluble cluster of differentiation 14, IL-18, and soluble tumor necrosis factor receptor 2.[20,21] As IL-18 and C-reactive protein are connected to the nucleotide-binding domain, leucine-rich repeat, and pyrin domain-containing protein 3 (NLRP3) inflammasome/IL-1β pathway, we sought to broaden our analysis to assess a large panel of inflammatory markers in the plasma proteome using the Olink Explore platform. We found that a large number of plasma inflammatory biomarkers, wound healing pathways, and the matrix remodeling protein TIMP3 predicted incident AMD in this setting, even after correcting for the FDR. Variants in the TIMP3 gene have been linked to AMD in HIV-uninfected populations,[4] and in vitro studies suggest that blocking TIMP3 may inhibit drusen formation, the hallmark of early- and intermediate-stage AMD.[29] Furthermore, the IL-1β pathway was highly enriched among the proteins predicting AMD. This relationship appeared to be specific to AMD as a far larger number of inflammatory pathways (including not only just innate but also activation of adaptive immune system pathways) predicted increased risk of mortality and a decreased risk of cataracts.

These data identify—using an unbiased, discovery-based proteomic approach—the IL-1β pathway as a significant predictor of and central to incident AMD. Activated IL-1β is released from cells after the proteolytic cleavage of pro-IL-1β by caspase-1, which in turn is induced by NLRP3 inflammasome activation.[30] The NLRP3 inflammasome and the downstream IL-1β signaling pathways are critical early events in the innate inflammatory response, triggered by pathogen-associated or damage-associated molecular patterns.[31] The NLRP3 inflammasome pathway has also been associated with AMD pathogenesis in both animal models and possibly in vivo.[32,33] The linked plasma marker IL-18, which also is released from cells after proteolytic cleavage of pro-IL-18 by NLRP3 inflammasome-induced caspase-1, has been associated with AMD in people with and without HIV infection.[22,34] Our finding that the IL-1β pathway is associated with incident AMD in a discovery-based plasma proteomic approach and identification of its central role further supports the body of evidence implicating this pathway in AMD pathogenesis. This identification of this pathway is important as direct inhibitors of IL-1β have been studied in clinical trials of treatments for age-related diseases, including the CANTOS Trial, where canakinumab reduced recurrent cardiovascular event rates and cancer mortality in HIV-uninfected persons.[35] The fact that TIMP3, the only protein with known genetic associations with AMD that was represented in the Olink Inflammation I panel, was also associated with increased AMD risk further underscores the robustness of this proteomics platform.[4]

Our observation that inflammatory markers tended to predict a decreased risk of cataracts in people with AIDS, even after accounting for the competing risk of mortality, also is consistent with our hypothesis-driven work from the same cohort.[21] We do not have a clear mechanistic explanation for these findings. Although we initially suspected that the competing risk of mortality in this cohort might have resulted in a survivorship bias, enriching for decreased inflammation among those who survive long enough to get diagnosed with cataract, this association persisted even after accounting for competing risk of mortality with sub-hazard ratios.[36,37] Additional studies will need to be done in people with less advanced HIV infection as well as in the general population to understand the generalizability of these findings. Lastly, our observation that a much larger array of inflammatory pathways predicts increased mortality in this setting—spanning not only just innate immune system activation but also adaptive (i.e., T and B cell) immune system activation—speaks to the specificity of our finding that the IL-1β pathway is associated with increased AMD risk in this setting.

Limitations

Our study also did not include HIV-uninfected persons, but there are immunologic similarities between immunosenescence in HIV-uninfected persons and the changes seen in ART-treated, immunorestored, HIV-infected persons, suggesting potential relevance for the HIV-infected population.[14,15] The demographics of our participants with AMD are reflective of the AIDS epidemic and are unusual for AMD in the general population.[18,22] Hence, additional studies need to be conducted in an HIV-uninfected population to test whether these observations can be translated to people without HIV infection. The LSOCA cohort did not have ocular specimens, so we cannot assess the intraocular levels of the plasma inflammatory biomarkers associated with AMD. Although the absence of intraocular data does not invalidate the significant associations found, we cannot distinguish between a systemic process affecting AMD pathogenesis indirectly or a direct effect within the eye. Lastly, the sample size of our incident AMD analysis is small, which limits our ability to simultaneously adjust for many potential confounders, but since the characteristics of our AMD cases and controls were well matched by ART status (there were no significant differences in ART use at baseline, and all participants and controls were ART-treated during follow-up with regimens including a nucleoside transcriptase inhibitor) and comorbidity prevalence (there were no significant differences between the 2 groups), significant confounding by these factors seems very unlikely.[21]

Conclusions

Using a discovery-based plasma proteomics approach, we identified a strong association between systemic inflammation and incident AMD in people with AIDS, a central role for the IL-1β pathway, and an association between elevated systemic levels of TIMP3 and incident AMD. If these associations also are present in an HIV-uninfected population, then the IL-1β pathway and/or TIMP3 may be potential targets for interventions to decrease the incidence of AMD.

Footnotes and Disclosures

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No animal subjects were used in this study.

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Analysis and interpretation: Hunt, Olshen, Murad, Sezgin, Schneider, Jabs

Obtained funding: Hunt, Jabs

Overall responsibility: Hunt, Olshen, Jabs

Abbreviations and Acronyms:
AMD = age-related macular degeneration; **ART** = antiretroviral therapy; **FDR** = false discovery rate; **HIV** = human immunodeficiency virus; **IL** = interleukin; **LSOCA** = Longitudinal Study of the Ocular Complications of AIDS; **NLRP3** = nucleotide-binding domain, leucine-rich repeat, and pyrin domain-containing protein 3; **TIMP3** = tissue inhibitor of metalloproteinase 3.

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